

Summary for 2024-10-28-Farkas

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1 Summary

Farkas' Lemma is a fundamental theorem in linear programming that establishes relationships between the feasibility of a system of linear inequalities and the existence of solutions to a related dual system. The lecture explored various formulations of this lemma, highlighting its versatility in analyzing linear programs.

* **Farkas' Lemma:** A system of linear inequalities, $Ax \leq b$, has no solution if and only if there exists a non-negative vector y such that $Ay = 0$ and $by < 0$.

* **Alternative Formulations:** The lemma can be expressed using equality constraints ($Ax = b$) and non-negativity constraints on x , leading to alternative conditions for solvability involving a dual system in y .

* **Three Variants:** The lecture presented three equivalent variants of Farkas' Lemma, each addressing the solvability of systems with different combinations of equality and inequality constraints, and non-negativity conditions on the solution vector x .

* **Connection to Fredholm Alternatives:** The lecture highlighted the relationship between Farkas' Lemma and the Fredholm Alternatives, demonstrating its broader mathematical significance beyond linear programming.

* **Conditions for Non-Existence of Solutions:** Several slides provided conditions under which the primal systems ($Ax \leq b$ or $Ax = b$) have no solution, expressed in terms of the existence of a solution to a corresponding dual system.

* **Geometric Interpretation:** The proof of Farkas' Lemma was outlined using a geometric approach, involving the projection of a point onto a cone defined by the matrix A .

* **Related Theorems:** The lecture included a problem requiring the derivation of several theorems (Gordan, Farkas, Stiemke, Ville, and Tucker) closely related to Farkas' Lemma, showcasing its wide-ranging applications in mathematical analysis.